RevertLab: Cost-aware continuation under rival spread dynamics

An auditable negative result in chronological pairs-trading research

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Abstract

A large spread is not itself an attractive trade. We evaluate a sequential rule that values a specified convergence policy under stable mean reversion and a persistent rival, propagates parameter uncertainty, charges both-leg execution and borrowing, and compares retaining actual shares with delayed liquidation. The empirical contribution is an auditable test of decision integration beyond a hard breakdown gate, with joint factor/spread scenarios and a separately implemented accountant. It is not a new optimal-stopping theorem. In a six-asset, three-pair crypto study with formation-only selection and an initially reserved 2023–2025 evaluation, the nested mixture rule loses 1.20% on deposited capital. Its annualized arithmetic mean-return improvement over the development-selected gate is 1.30 percentage points, with a paired 95% interval of $[-3.17, 5.54]$ points and an expanded-family adjusted $p = 0.715$. A development-scaled cointegration control earns 3.31% with less drawdown. Known-process tests reveal that volatility changes masquerade as relationship failure. These results do not establish an economic advantage. All decisions, rejected opportunities, physical fills and cashflows are retained for reproduction and challenge.

1 Question, contribution and competition

The question is whether a position remains worthwhile given its convergence timing, rival dynamics, costs and downside, rather than whether two histories are correlated. The proposed mechanism combines a forward-only regime filter with first-passage valuation of actual fixed quantities. Its falsifiable contribution is the measured incremental value of that integration against the *same predictor used only as a gate*, stable-only value, adaptive hedge rules, cash and a reduced-exposure control. The measured outcome does not support a positive-advantage interpretation in this sample.

The ingredients have substantial precedents. Gatev, Goetzmann and Rouwenhorst study normalized-distance selection [1]. Song and Zhang formulate cost-sensitive entry/exit with stop loss [2]; Ngo and Pham study optimal switching across flat and both pair directions [3]. Altay, Colaneri and Eksi combine hidden-drift filtering and risk penalization [4]. ArbitrageLab implements distance, cointegration, Kalman, OU entry/liquidation and AR(1) minimum-profit timing methods [6]. First-passage time, regime filtering and an economic stopping threshold therefore cannot be claimed as inventions here. Dou et al.'s September 2026 work directly monitors when to pause and resume pairs trading [5]; only its author abstract was accessible, with full text returning HTTP 403. Its exact procedure was not reproduced, so superiority to it is untested.

We compared the formulations of supervised competing-risk hazards, filtered rival-dynamics valuation and conservative stable-only valuation before choosing the second. Supervised hazards remained an unimplemented alternative; the small formation-only estimator did not require ML. This work supplies a transparent empirical mechanism and falsification package, not a globally optimal partially observed control policy or a universal classifier of economic breakdown.

2 Data and chronological protocol

Six surviving spot instruments were fixed before strategy returns: BTC, LTC, ETH, ETC, XRP and XLM, quoted in USDT on Binance. Three candidates were fixed by economic relationship: BTC/LTC (proof-of-work monetary

assets), ETH/ETC (shared chain history), and XRP/XLM (payment networks). These motivations do not imply equilibrium or market neutrality. Retrospective survivor selection is explicit; point-in-time membership was unavailable.

The retained panel covers 1 January 2019 to 31 December 2025: 2,557 complete UTC days per asset, 15,342 rows, and 504 monthly archives. Every ZIP was checked against its official SHA-256 file. Ingestion validates schema, positive finite OHLC/volume, high/low ordering, duplicate/gap absence, UTC interval boundaries, and the switch from millisecond to microsecond timestamps in January 2025. Offline acquisition replay reproduced the normalized hash. Binance Vision's explicit CC BY-NC-SA 4.0 terms permit the retained noncommercial research route [11]; code licensing is separate. Latest-vintage archives do not establish original publication vintages or revision histories. USDT is not USD.

A close in day d becomes available at the boundary starting $d+1$. A voluntary decision then fills at the open of $d+2$, strictly later than its information. The delay stress fills at $d+3$. Forecast latency is correspondingly one or two daily transitions from the observed close. No intraday stop ordering or guaranteed stop price is inferred from bars.

Formation uses calendar 2019, development 2020–2022, and the initially reserved final interval 2023–2025. Every six-month outer fold uses its preceding 365 days. The first 183 days of that formation window estimate/select an inner pair model; the remaining days evaluate the eleven registered configurations. Inner choice maximizes mean daily capital return minus half its daily standard deviation. The outer model re-estimates on all 365 formation days and ranks the predefined candidates by Engle–Granger p-value, admitting $0 < \hat{\phi} < 1$ and $0.1 \le \hat{\beta} \le 4$. At most two candidates trade. This is a ranking rule, *not* a claim of significant cointegration after selection. Distance separately ranks base-price-normalized squared distance without stationarity admission.

Training has no forward outcome labels. Inner trades are liquidated within their evaluation boundary, so their outcomes do not overlap the outer test. Later final folds may use earlier final observations only under the prespecified rolling algorithm. No future-conditioned smoothing is used. The final interval was opened only after development, data/code/config hashes and tests were frozen. It is now observed. A later metadata-only control-summary repair was reproduced without changing any economic paths; that reproduction is not another unseen evaluation. Local hashes are not external timestamps and do not reveal trials outside this repository.

3 Model, physical positions and decision value

For formation-frozen log-price regression, let

$$x_t = \frac{\log P_t^A - \hat{\alpha} - \hat{\beta} \log P_t^B - \hat{\mu}}{\hat{\sigma}}. \tag{1}$$

The center and residual standard deviation use formation data only. The stable predictor is $x_{t+1} = \phi x_t + \epsilon_{t+1}$; the rival is $x_{t+1} = x_t + 1.5\epsilon_{t+1}$. The online filter uses Student-t_5 predictive densities with innovation variance matched to formation residuals. If p_t is the rival weight,

$$p_{t+1}^- = p_t(1-r) + (1-p_t)h, \tag{2}$$

$$p_{t+1} = \frac{p_{t+1}^- f_B(x_{t+1} \mid x_t)}{p_{t+1}^- f_B + (1-p_{t+1}^-) f_S}, \qquad h = 1/365, \quad r = 1/90. \tag{3}$$

The state is a conditional rival weight, not calibrated structural-break truth. The small recovery transition prevents an old alarm from permanently banning the pair. Inspection of the constant-hazard BOCD reference showed why its reset probability alone can be uninformative [8, 7]; we did not rename that scalar as a detector.

For $d = -\operatorname{sign}(x_t)$ and gross budget G, actual entry quantities are fixed at the decision:

$$q_A = \frac{dG}{(1+\hat{\beta})P_t^A}, \qquad q_B = -\frac{\hat{\beta} q_A P_t^A}{P_t^B}. \tag{4}$$

Prices at the later fill are unknown when these quantities are set. A log coefficient is never booked as a share quantity or a profit. Static/value books retain actual shares until exit. Rolling and Kalman baselines queue and charge every physical rehedge.

The specified forecast policy observes closes, triggering on convergence $dx \geq 0$, adverse movement $d(x - x_{\text{signal}}) \leq -4$, or age 20 days, then fills after the same latency. At fold ends all positions close at the preannounced final open. New entries are withheld within $20 + 2D + 2$ bars of that boundary, where $D = 2$ in the base experiment. The adverse event is a residual trading barrier, not proof of relationship failure. Forecast convergence/adverse/expiry probabilities sum to one; delayed realized prices can overshoot barriers. Actual value decisions re-evaluate daily, so their realized holding times need not match this specified forecast policy.

We sample paired formation innovations and mean-centered B-leg log returns in five-day blocks. For each path, with $b_u = \log(P_u^B / P_t^B)$,

$$P_u^B = P_t^B e^{b_u}, \qquad P_u^A = P_t^A \exp\{\hat{\beta} b_u + \hat{\sigma}(x_u - x_t)\}. \tag{5}$$

This retains formation correlation and the convexity of actual fixed-share holdings, unlike valuing only log-spread movement. Twelve parameter ensembles vary ϕ by its formation regression standard error and the innovation scale by a chi-square variance draw; ϕ is clipped to $[0, 1.02]$, retaining nearly unit-root and mildly explosive alternatives. Each ensemble has 48 paths. Hedge and transition uncertainty are not integrated. Student-t filtering and empirical-bootstrap forecasting are deliberately different approximations, not a single exact posterior model.

For entry, path gain $g_{e,j}$ is physical both-leg value change between delayed entry and delayed exit, less each path's turnover costs and signed borrowing, divided by decision-time gross budget. For an existing book, current actual signed leg fractions replace entry fractions and

$$g_{e,j} = G_{e,j}^{\text{continue}} - G_{e,j}^{\text{liquidate after same delay}}. \tag{6}$$

Forecast friction uses reference notional, omitting the small fee-times-impact cross term that the actual ledger charges on adverse fills. Shared liquidation costs cancel except for their path-dependent notional differences; sunk entry costs are not charged again. Additional borrowing is charged only while continuation outlasts liquidation.

Let $\bar{g}_e$ be the mean within ensemble e, and let $L_{.05}$ be the nonnegative mean loss in the worst 5% of pooled *net* outcomes. The score is

$$J_t = Q_{.10}(\bar{g}_1, \ldots, \bar{g}_{12}) - \lambda L_{.05}. \tag{7}$$

An entry requires $|x_t| \geq 2$ and $J_t > 0$; continuation requires $J_t > 0$ unless a barrier/horizon/boundary mandates exit. Registered mixture candidates use mean only, ensemble quantile alone, or quantile plus $\lambda \in \{0.25, 0.50\}$. The displayed score applies to uncertainty-aware variants. The mean-only variant instead uses the pooled net mean, omits parameter draws, and sets $\lambda = 0$. The nested mixture chooses inside each formation window. The quantile is a conservative heuristic, *not a calibrated lower confidence bound*; Monte Carlo standard error clusters across ensembles. This local book downside does not model the full shared-capital margin or borrow-availability process. The model is fixed-policy Monte Carlo valuation with receding-horizon action selection, not a Bellman solution.

4 Accountant and execution safeguards

Deposited capital is 100,000 USDT per evaluated method. Equity is $E = C + \sum_a q_a P_a$; returns use successive marked equity including initial cash. For a signed fill Δq at open reference P, fill price is $P[1 + \text{sign}(\Delta q) 5 \cdot 10^{-4}]$ (2 bp half-spread plus 3 bp slippage); commission is 5 bp of executed notional. Borrow is 5% annually, debit financing 8%, cash yield zero, ACT/365 on beginning open marks over the elapsed open-to-open interval. Short collateral is 1.02 times marked physical short liability; financing uses cash less that restricted amount. Short proceeds do not expand deposited equity.

Base targets face gross/equity ≤ 1.5, absolute net/equity ≤ 0.4, individual asset/equity ≤ 0.6, book/equity ≤ 0.5, and 50% initial margin. Total pair-book gross is also capped, so offsetting books cannot hide large exposure. Each entry requests 35% of current equity, further capped per leg at 0.1% of the preceding 20 completed days' median quote volume. That volume is a scale proxy, not opening liquidity. Actual positions may drift; equity below 30% of gross triggers forced liquidation. A margin gap is marked and forcibly filled at the observed open with adverse costs, an explicit same-instant emergency approximation rather than invented earlier information. This branch is verified by hand cases; it was not exercised by the real study.

Borrow availability is checked against *aggregate physical shorts*; reducing an offsetting long can require new borrow. The base two-leg fill is conditional atomic execution after preflight. A deterministic 10% entry-attempt subset in the asymmetric stress fills one leg and charges immediate adverse rollback, assuming the unwind is available. There are no guaranteed intrabar stops, historical opening-depth claims or recovered historical borrow records.

Signed dividends, splits, borrowing and financing are supported and hand-checked. These spot paths have no ordinary equity actions; ETH/ETC positions are excluded from 1 September through 30 September 2022, with prior flattening, around the preannounced Merge [12]. Unsupported minor fork/airdrop rights, staking, taxes and exchange default are outside scope. No execution adjusts prices while also booking the same action.

A separate event accountant reconstructs inventory, fill prices, fees, carry and marked equity without invoking the trading accountant's calculations. Across 354 outer fold files, 33,902 events, 64,664 snapshots and 2,001 trades, cash errors are zero, maximum equity discrepancy is 2.91×10^{-11} USDT, and maximum completed-trade attribution discrepancy is 2.02×10^{-10}. Merged daily returns match their source folds exactly. All 13,563 saved value scores match their recorded formulas/signs. These checks establish internal numerical consistency, not real-market fill attainability.

5 Comparisons and results

All methods use the same bars, capital, delayed-fill accountant, costs and limits. Distance is a Gatev-style base-price-normalized variant, not exact ArbitrageLab min/max normalization. Cointegration-ranked uses frozen positive log hedges and threshold exits. Rolling refits on the latest 60 observations; Kalman uses one-step prediction errors before measurement updates and books rehedges. The same rival filter gates threshold entry/holding at a nested weight threshold of 0.2 or 0.5. Stable-only value removes the broken rival but retains uncertainty and the 0.25 downside penalty. These are substantial competitors, although exact OU optimal boundaries and the recent Dou procedure remain unimplemented.

Method	Dev. return %	Final return %	Final DD %	Mean gross % equity	Trades	Turnover × capital
Distance	18.09	-1.22	-14.35	19.01	40	21.7
Cointegration-ranked	-9.07	6.03	-16.34	20.27	41	23.0
Rolling hedge	-59.22	-3.02	-12.96	17.41	44	24.9
Kalman hedge	-9.98	8.60	-2.94	1.68	6	3.1
Breakdown gate	7.03	-5.22	-8.24	8.56	31	16.4
Stable-only value	20.89	-0.05	-14.51	13.82	78	42.8
Nested mixture value	0.27	-1.20	-4.70	2.72	34	20.2
Scaled cointegration	-2.60	3.31	-1.99	3.37	42	3.7
Cash (zero yield)	0.00	0.00	0.00	0.00	0	0.0

Table 1: Net capital results. Development and final intervals are each three years. Drawdown includes initial capital. The reduced control's scale 0.1193027774 was frozen from development; achieved exposure is approximate, not exact final matching. Cash has stated zero yield.

Development selected the hard breakdown gate as the strongest baseline by the frozen mean-minus-volatility criterion, not by final returns. On final data the mixture loses 1,198.81 USDT in 34 round trips, paying 1,008.41 in fees, 1,008.41 in adverse fill friction and 267.02 in borrow. Mean holding is 2.91 days, and 538 potential entries are rejected. None of its 34 exits is labeled a convergence-trigger exit; most are continuation-value exits. Thus it should not be sold as a demonstrated convergence-capture strategy.

Against the frozen gate, the annualized *arithmetic mean daily return* difference is +1.305 percentage points, with paired moving-block 95% interval $[-3.175, 5.537]$. This is not an interval for CAGR, compounded return difference, or future profitability. Circular 20-day blocks and 999 draws preserve aligned dependence across methods. The one-sided $p = 0.272$ is unadjusted; max-t adjustment across seven families gives 0.601, and across those plus four pre-registered fixed ablations gives 0.715. Hansen SPA via `arch.bootstrap.SPA`, using negative daily capital returns as loss and a stationary 20-day bootstrap, gives consistent global $p = 0.417$ (0.440 expanded) [10]. These global tests concern whether any supplied model outperforms the benchmark, not the mixture alone. Block sizes 5 and 60 are retained as sensitivity analyses. All intervals condition on fitted paths; they do not refit every selection step or correct survivor-universe choice.

The reduced cointegration control earns 3.31% with 1.99% maximum drawdown and 3.37% average gross, versus the mixture's 2.72% gross. Kalman earns 8.60% with 1.68% gross, but only six trades. These observations contradict attributing the mixture's lower drawdown to uniquely intelligent breakdown handling; neither establishes

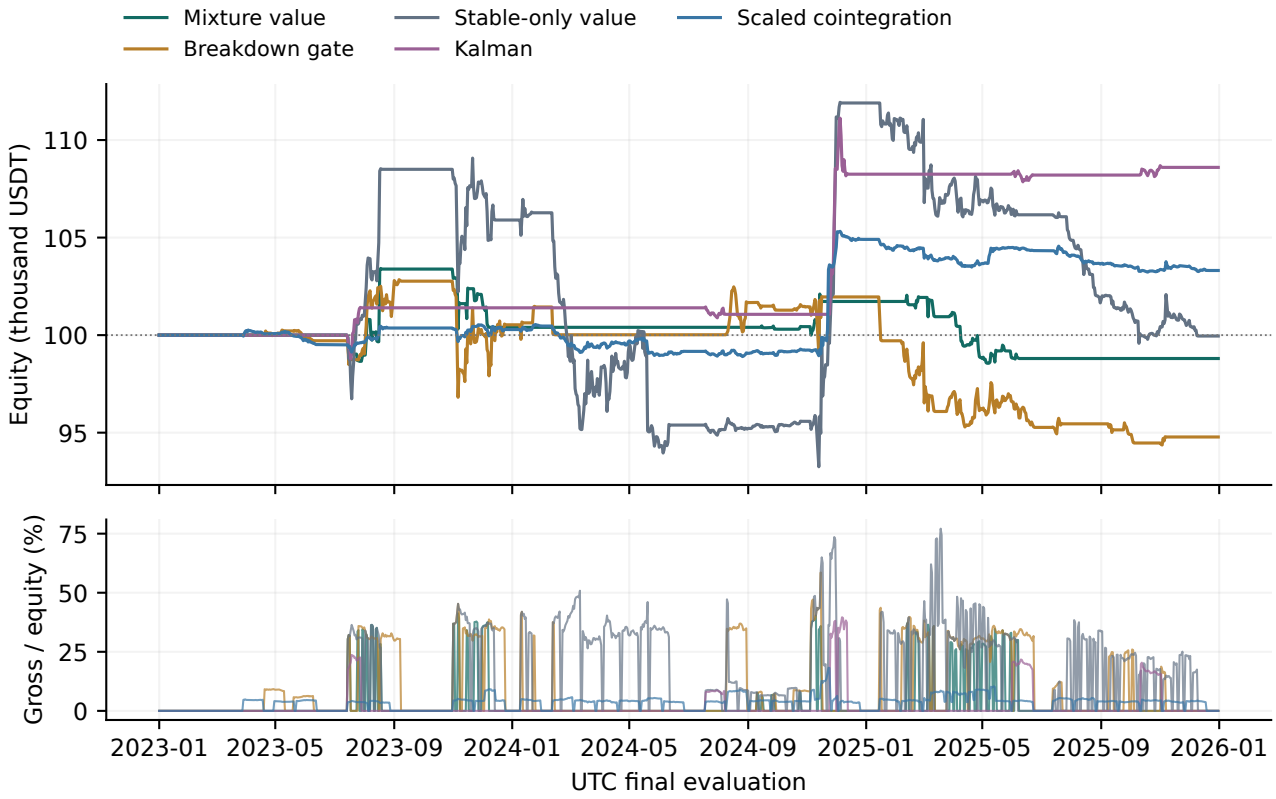


Figure 1: Actual saved ledger equity and gross exposure during the final interval. Blank activity is genuine flat exposure. Smaller drawdown alone cannot establish a better detector when capital use differs.

a generally profitable alternative. Aggregate pair/year attribution and exact dates are in the saved paper-values file and terminal.

Fixed mixture ablation	Return %	DD %	Gross %	Trades	Adjusted p
Mean, no ensemble penalty	5.93	-17.41	18.80	46	0.766
Lower ensemble quantile	-7.05	-14.63	13.19	67	1.000
Quantile + tail penalty 0.25	0.88	-5.82	4.67	40	0.468
Quantile + tail penalty 0.50	-1.59	-5.08	2.58	31	0.752

Table 2: Unchanged configurations registered before returns, evaluated as secondary ablations. Adjusted one-sided max-t probabilities use all eleven paths against the development-selected gate. No final-period winner is promoted as a new discovery.

The ablations do not establish that uncertainty treatment improves net performance. The mean-only variant earns 5.93%, while the ensemble-quantile version loses 7.05%. The 0.25 downside penalty earns 0.88%, and 0.50 loses 1.59%. These observed differences are sensitive, selected-path descriptions; their uncertainty does not support choosing a new final-period optimum.

Doubling execution costs worsens the mixture to −3.31%; 25% borrowing to −7.58%. An additional daily-bar delay changes it to +2.34% and also materially changes competitors. Borrow unavailable leaves every strategy flat, as required. Failed-leg attempts incur unwind costs; their apparently improved mixture endpoint reflects different sampled trades, not a benefit from execution failure.

Final-period stress	Mixture return %	Mixture DD %	Gate return %	Stable-only return %
Base assumptions	-1.20	-4.70	-5.22	-0.05
Double execution costs	-3.31	-6.10	-6.80	-3.28
Extra daily-bar delay	2.34	-6.69	3.73	3.25
25% annual borrow	-7.58	-10.22	-8.28	1.67
All borrow unavailable	0.00	0.00	0.00	0.00
10% failed-leg subset	-0.49	-3.86	-7.02	0.03

Table 3: Frozen selections, recomputed quantities and cashflows. Cost/borrow-aware rules may change decisions under the stressed terms. Favorable delay or rollback outcomes are sampling/path effects, not guaranteed protection.

6 Known-process falsification and calibration

We simulate 96 replications per regime with known formation parameters, a temporary innovation, and held-out monitoring. Cases are stable AR(1), a 120-day persistent random-walk/drift segment followed by recovery, mean reversion around a shifted equilibrium, and doubled innovation volatility with the original mean reversion intact. These are detector/calibration diagnostics, not market profitability tests or tuning targets. The persistent-break and shifted-mean cases both generate a five-consecutive-day alarm in every replication, with median delays 12 and 9.5 days respectively. Detecting a shifted mean is not the same as detecting loss of mean reversion.

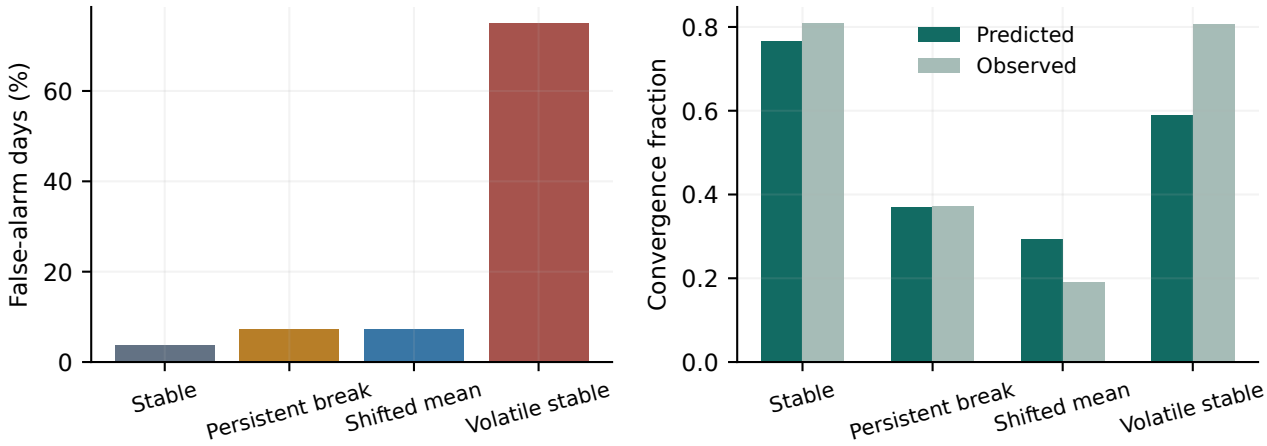


Figure 2: Known-process failure. False alarms are the fraction of stable-truth evaluation *days* for which rival weight exceeds 0.5, not a fraction of replications. Forecast convergence uses separate nonoverlapping evaluation windows.

The stable case has 3.71% false-alarm days, but volatile stable behavior has 74.84%. The model therefore confuses volatility with breakdown. Its volatile-stable convergence forecast averages 58.95%, versus observed 80.60%; its mean-shift forecast averages 29.24%, versus 19.01% observed. Robust generalized-Bayesian change detection is a relevant future competitor [9], not a capability demonstrated here.

On actual selected, nonoverlapping 24-day opportunities, final calibration has only 38 observations and one convergence, with Brier score 0.01569. Development has 38 opportunities and no convergence. Such event scarcity cannot establish calibrated convergence probabilities; low scores can reflect low base rates. The numerical diagnostic agrees on score sign in twelve formation-only representative states across 576, 2,304 and 6,144 paths, using one pair and seed. That limited check does not establish global decision stability or validate the parameter quantile.

7 Reproduction, limits and conclusion

The offline terminal links each trade or rejected opportunity to what was known, its formation model, forecast probabilities/timing, physical targets, both legs, rehedges, carry and realized cashflows. Its charts use the same ledger snapshots as metrics and exports. Source, configuration, trial registry, raw results, data-access script, hashes, independent checks and compiled-paper sources are retained. The original initially-unseen evaluation is preserved; the metadata-corrected reproduction has identical economic metrics. Performance figures come from the retained historical simulation; no live trading was evaluated.

The main limits are the retrospectively chosen survivor universe, three economically restricted candidates, latest-vintage bars, hypothetical spot shorts, daily execution/depth assumptions, uncertain fork/venue risks, small independent regime count and conditional bootstrap inference. Model probabilities are not structural truth; transition/hedge uncertainty and shared-capital tail dependence are incomplete. Dollar hedges do not establish market neutrality. The terminal makes these assumptions inspectable rather than resolving them.

The proposed integration did not earn a demonstrated out-of-sample net advantage. Its useful quantitative result is the combination of a reproducible economic test and an explicit falsification: conservative mixture scoring reduced activity, but exposure controls and volatility-induced alarms undermine an intelligence claim. Future work should first distinguish volatility change from loss of reversion and validate convergence timing on genuinely new data. Reusing this final period to choose a favorable variant would not answer that question.

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